Technical Summary Sheet — AI Express Hackathon

Autonomous Rescue Bot: Dynamic A* Replanning Agent in a Partially Observable Grid

1. HEADER & TEAM INFO

Course Code	BCA 301-5 — Artificial Intelligence	**Group ID**	**<GROUP ID>**
Selected Track	Unit 2 — Problem-Solving Agents (Informed Search + Dynamic Replanning)	**Date**	**<DD MMM 2026>**
Team Members	**1. <Name> (<Reg No>) 2. <Name> (<Reg No>) 3. <Name> (<Reg No>)**		
GitHub Repository	**https://github.com/<org>/<repo>**		
Demo Video	**<video link> (also embedded in README.md)**		

2. PEAS FRAMEWORK MATRIX

Performance Measure	Goal reached (boolean); total path cost g(goal) in unit moves; number of nodes expanded per plan; number of replans triggered; wall-clock time to goal; collisions = 0 (hard constraint).
Environment	15×15 discrete occupancy grid with static obstacles unknown at t=0. **Partially observable** (sensor radius $r = 2$), **deterministic** transitions, **sequential**, **dynamic from the agent's viewpoint** (belief state changes as cells are revealed), **discrete**, **single-agent**, **known** transition model.
Actuators	Four deterministic movement primitives — MOVE_UP, MOVE_DOWN, MOVE_LEFT, MOVE_RIGHT (4-connected, unit cost). One cell per simulation tick; blocked moves are rejected before execution.
Sensors	Local proximity scan returning the occupancy status of every cell within Chebyshev distance $r = 2$ of the agent; internal odometer reporting current position (x, y); goal coordinate register.

3. CORE ALGORITHMIC FORMULATION

State Space	$S = \{ (x, y) \mid 0 \le x, y < N \} \setminus B_{known}$, where B_{known} is the set of obstacle cells revealed so far. $\lvert S\rvert \le N^2 = 225$. The agent plans over its *belief grid*: unobserved cells are optimistically assumed free (free-space assumption), which is what makes replanning necessary.
Initial State	$s_0 = (0, 0)$, belief grid $B_{known} = \varnothing$, $g(s_0) = 0$.
Goal Test	$\mathrm{GoalTest}(s) \equiv (s = s_g)$, with $s_g = (14, 14)$. Single-state goal, tested on node expansion.
Transition / Path Cost	RESULT(s, a) applies a unit displacement if the target cell is in-bounds and not in B_{known}. Step cost $c(s, a, s') = 1$ for all legal moves, so $g(n) = \mathrm{depth}(n)$ and total path cost = number of moves executed.
Evaluation Function	`f(n) = g(n) + h(n)`
Heuristic	$h(n) = \lvert x_n - x_g\rvert + \lvert y_n - y_g\rvert$ (Manhattan / L1 distance) **Admissible:** with 4-connected unit-cost moves, no path can close one unit of L1 distance in fewer than one move, so $h(n) \le h^*(n)$ for every n. **Consistent:** for any neighbour n', $h(n) \le c(n, a, n') + h(n')$ since L1 changes by exactly ±1 per move. Consistency guarantees A* returns an optimal path on the current belief grid without node re-expansion. *Note:* Manhattan becomes inadmissible if diagonal moves are enabled — octile distance would be required.
Replanning Rule	After each SENSE step, if any cell on the remaining planned path $\pi = \langle n_k \dots n_{goal}\rangle$ is found in B_{known}, the plan is invalidated and A* is re-executed from the agent's current cell. This is the LRTA*-style sense–plan–act loop: `while not at goal: sense(); if path_blocked(): replan(); step()`. If OPEN empties before the goal is expanded, the goal is unreachable and the agent halts.

4. COMPLEXITY ANALYSIS

Aspect	Theoretical (O notation)	Justification	Observed
Time — single A* plan	$O(b^d)$ worst case; $O(\lvert E\rvert \log \lvert V\rvert) = O(4N^2 \log N^2)$ on a grid	b = 4, d = solution depth. Binary-heap OPEN gives log-factor pops; each of the N^2 cells has ≤ 4 edges.	**<__> nodes expanded**
Space — single A* plan	$O(b^d)$, bounded by $O(N^2) = O(225)$	A* stores every generated node; on a finite grid, OPEN ∪ CLOSED can never exceed the cell count.	**<__> peak frontier size**
Time — full mission (k replans)	$O(k \cdot N^2 \log N)$	Each obstacle discovery triggers at most one replan; k is bounded by the number of revealed blocking cells.	**k = <__> replans** **<__> s wall clock**
Sensing overhead per tick	$O(r^2) = O(25)$, constant	Fixed $(2r+1)\times(2r+1)$ scan window, independent of grid size.	negligible
Solution quality	Optimal per replan; bounded-suboptimal overall	Consistent h guarantees optimality on the belief grid, but the free-space assumption means the executed trajectory may exceed the true optimal path.	**cost <__> vs. oracle <__>**

Observed vs. theoretical: measured expansions stayed well below the b^d worst case because the consistent Manhattan heuristic keeps A* strongly goal-directed — the frontier grows as a narrow corridor rather than a uniform disc. Replans are progressively cheaper as the agent nears the goal (smaller residual d). **Replace all <__> fields with the figures printed by logger.py at the end of the run.**